AI Predicts, ArcGIS Pro Proves: Checking Planning Forecasts Against Reality

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Title	AI Predicts, ArcGIS Pro Proves: Checking Planning Forecasts Against Reality (75/80)
Abstract	AI can forecast in seconds where a city will build, but it does not yet understand the physical world: real parcels, real streets, years of construction. Its maps look right and can be wrong. We use ArcGIS Pro as the reality check. A planning AI from six years of PhD research forecast redevelopment on 42,000 Manhattan lots in 2020. Pro scored it against the permits and certificates of occupancy issued since. The AI beat a planners' rule of thumb on finished buildings, but not on construction starts. Planners and GIS analysts leave with the workflow, a checklist and a game to try. 586/600 characters. Published as written in the agenda.
Keywords	AI validation; ArcGIS Pro; urban planning; redevelopment forecasting; public records
Topic category	State and local government: Planning (focus area: Efficiencies and Innovations in Planning)
Audience and level	Planners, GIS analysts and anyone using AI predictions for places. Intermediate; no AI background needed.
Primary speaker	Parya Payami, Department of Architecture, University of Wisconsin-Milwaukee (six years of PhD research in urban planning, AI and GIS)

Expanded abstract

The problem. AI models now produce planning forecasts, impact estimates and coverage maps in seconds. They learn patterns from data, but they do not yet understand the physical world: where a building really stands, how long it takes to build, how fast a truck can drive on a real street. Their output looks as polished as a correct answer, so errors pass unseen.

The missing piece. Cities already publish what really happened: permits, certificates of occupancy, dispatch times. ArcGIS Pro is where a prediction and those records meet on the same ground, parcel by parcel and street by street. The AI predicts; Pro checks it against reality and shows where it missed.

Reality the AI misses	What we found	How ArcGIS Pro handles it
Location	Certificates of occupancy are placed on the street, not the lot: only 30 of 601 fell inside a lot, so a plain overlay counted 16 finished lots instead of 430	Join by tax lot (BBL), then Generate Near Table for the rest; an automatic warning flags the problem
Time	Buildings finish years after they start: 53% of lots certified after 2020 already had a recent building that year	Definition queries score only lots that could still change
Capacity	Three Manhattan rezonings allowed 8,423 more homes; 561 were built by 2025	Capacity is mapped as an end state, never as a forecast
Streets	Drive times at posted speeds were a median 2.2 min faster than FDNY's recorded travel times (167 areas)	Network Analyst on a network built from city streets, corrected by observed times (92% within 1 min, leave-one-out)

Main case: did Manhattan build where the AI said it would? The AI is PlanArena, a parcel-level planning model from our PhD research (4.99 million parcels, ten US cities). Its 2020 forecast used only data available in 2020; the records came later and were never used to build it. The rule of thumb is the planners' soft-site rule (lots built to well under their zoning).

Test (2020 forecast, 42,170 lots)	Observed records, 2021-2025	AI forecast	Planners' rule of thumb	What it shows
Where construction started	DOB NOW new-building first permits (150 lots)	AUC 0.81; top 10% of lots had 5.6x the base rate	AUC 0.81	The AI ranks starts well, but no better than the simple rule
Where buildings were finished	Certificates of occupancy for new buildings (215 lots)	AUC 0.67	AUC 0.39, worse than chance	The simple rule fails; the AI holds

Read honestly. The test checks where, not how many: ranking lots well is not predicting totals. Agreement with past records is not cause and effect, and the records have their own gaps, which each source documents.

Try it. The same AI runs as a browser game: paint land uses onto real Manhattan parcels and watch homes, jobs, emissions and access change. Play at planarena-manhattan.onrender.com/play (free server: the first visit can take a minute to wake).

AI agents too. We gave the same workflow to AI coding agents: 32 tasks, a fresh agent each, no access to the answers. All met pre-registered checks. Of 10 requests designed to mislead, they refused 3, redirected 5 and warned on 2, and they found 6 flaws in our own analysis, since fixed. One run per task, graded by the author.

Fifteen-minute plan.

Minutes	Content
0-2	One Manhattan district: what the AI forecast, what zoning allowed, what was actually built
2-5	The audience plays one scenario on their phones and sees the AI's impact estimates
5-11	Live in ArcGIS Pro: match permits and certificates to parcels, see the location and timing traps change the score, map the misses
11-13	The same check on streets: modeled drive times against recorded fire travel times; an AI agent runs the check
13-15	The one-page checklist for testing any AI prediction against reality
15-20	Questions

Takeaways. A tested ArcGIS Pro workflow for scoring an AI prediction against later public records, notes on 16 public outcome sources and their traps, and a one-page checklist. Code and results: github.com/ParyaPayami/proveit. Data: NYC MapPLUTO, DOB NOW filings and certificates of occupancy, FDNY dispatch; PlanArena manuscript (Payami, 2026).